A Small World Model at the 2026 Frontier: Discrete-Latent Imagination, Latent Actions, and Joint-Embedding Prediction, Built and Benchmarked From Scratch on a Laptop

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Abstract

A *world model* learns an internal, predictive representation of an environment and rolls it forward under actions. The 2018 Ha & Schmidhuber *World Models* system—a convolutional VAE (**V**ision), a mixture-density RNN (**M**emory), and a controller evolved inside its own "dream" (**C**ontroller)—remains the legible skeleton on which the 2024–2026 frontier is hung. We rebuild that skeleton with the techniques that define the current frontier and benchmark the two head-to-head on the same task, training *everything from scratch on a laptop GPU* (Apple MPS), no reinforcement-learning libraries. Our model, WM2, replaces the continuous VAE+MDN-RNN with a **discrete-latent Recurrent State-Space Model** (the DreamerV3 core: a deterministic GRU path plus a categorical stochastic state, trained with KL balancing, free bits, symlog targets, and two-hot reward/value heads), adds a **latent action model** (Genie-style, label-free) and a **decoder-free joint-embedding predictor** (V-JEPA-style), and replaces the evolved controller with **short-horizon model-predictive control through the model's imagination**. On a momentum-physics goal-reaching task the learned latent is a faithful *state*: a linear-ish probe recovers agent position at $R^2 = 0.96$ and—unlike the baseline, which never measured it—*velocity* at $R^2 = 0.55$. A controller that never touches the environment reaches the goal ∼**90%** of the time versus 25% for random, exceeding the 2018 baseline's 39%. We also report honest negative and partial results: open-loop dreams drift; a purely *learned*-reward actor-critic underperforms the probe-guided planner; and label-free action discovery only *partially* recovers the true actions (0.37 cluster purity) even though a supervised probe shows the action is 87% decodable from the latent transition—a precise, local instance of the unsupervised-action-learning open problem. We document the bugs we fixed along the way (foreground-weighted reconstruction, two-hot support scaling, and a 700× data-loading speedup) as practical guidance for building small world models. Code, artifacts, and a from-zero guide accompany this paper.

1 Introduction

A world model is a machine-learning system that builds an internal representation of an environment and predicts how that environment evolves under actions [1, 7]. The idea predates deep learning—Craik (1943) argued the mind runs a "small-scale model" of reality—but 2024–2026 is its inflection point, as reinforcement learning, computer vision, robotics, and generative AI converged on building such systems at once. By mid-2026 the term is, in Fei-Fei Li's words, "one of the most important and most overloaded terms in AI" [8].

The field has bifurcated into two paradigms. *Generative* interactive world models synthesize navigable environments frame-by-frame: DeepMind's Genie 3 [5] generates 720p worlds at 24 fps with $\sim$1 minute of visual memory, and Decart's MirageLSD streams real-time autoregressive-diffusion video at under 40 ms/frame. *Non-generative* predictors forecast in a learned latent space for understanding and control: Meta's V-JEPA 2 [6] trains a >1B-parameter video encoder with masked feature prediction and, with an action-conditioned predictor and model-predictive control, achieves zero-shot robot manipulation. In parallel, DreamerV3 [3] established a single discrete-latent model-based-RL recipe mastering 150+ tasks, and World Labs ships persistent 3D worlds (Marble). The common substrate of the *controllable* frontier is the Genie three-module design—a video tokenizer, a **latent action model**, and an autoregressive dynamics model.

These systems are enormous. We ask a complementary, pedagogical question: *what does the 2026 frontier look like at laptop scale, and how much does it actually buy over the 2018 baseline on the same task?* We build WM2, a small but architecturally faithful realization of the frontier's key ideas, and benchmark it head-to-head against a from-scratch reproduction of Ha & Schmidhuber's V-M-C model on an identical interface. Our contributions:

- A complete, from-scratch, laptop-trainable world model integrating four 2026 paradigms—discrete-latent RSSM (DreamerV3), latent action model (Genie), decoder-free joint-embedding prediction (V-JEPA), and latent model-predictive control (PlaNet/TD-MPC2)—with a verified numerical core (symlog, two-hot, straight-through categoricals).
- A controlled head-to-head against the 2018 baseline showing the 2026 model is a strictly more faithful *state* (it encodes velocity, not just position) and a better controller (97.5% vs. 39% goal-reaching), trained without ever touching the environment.
- Honest negative/partial results and a debugging narrative: the precise failure modes (long-horizon drift, learned-reward weakness, and a 87%-supervised vs. 37%-unsupervised gap in label-free action discovery) that mirror the field's open problems at toy scale.

2 Background and Related Work

The V-M-C lineage. Ha & Schmidhuber [1] factor an agent into Vision (a ConvVAE compressing frames to a latent z), Memory (an MDN-RNN modeling $p(z_{t+1} \mid z_t, a_t)$ as a mixture of Gaussians), and a tiny linear Controller evolved by CMA-ES *inside the model's dream*. PlaNet introduced the Recurrent State-Space Model (RSSM), splitting the latent into a deterministic recurrent path and a stochastic path; Dreamer [2] learns an actor-critic in imagination; DreamerV3 [3] makes the stochastic latent *categorical* and adds a robustness toolkit (symlog, two-hot distributional value/reward, KL balancing, free bits) that lets one configuration span 150+ tasks.

Generative interactive models and latent actions. Genie [4] learns to generate playable environments from unlabeled video via three modules, the central one being a **latent action model** that infers a discrete action between consecutive frames *with no action labels*. This is the canonical recipe for label-free controllability, and the module we reproduce.

The non-generative bet. LeCun's JEPA program [7] predicts in a learned embedding space rather than reconstructing pixels; V-JEPA 2 [6] operationalizes it for video and robot control via masked feature prediction against an EMA target encoder, followed by model-predictive control. We include a small decoder-free JEPA track to engage this paradigm.

The functional taxonomy. World Labs [8] classifies world models by output—*Renderers* (pixels), *Simulators* (state one can compute on), *Planners* (actions)—and argues these are "three projections of a single underlying understanding." WM2 is a miniature of exactly that: one learned latent that is rendered (decoder), simulated (RSSM dynamics), and planned over (MPC).

3 Method

3.1 Environment

We use a from-scratch 2D physics world (64×64×3 frames, 5 discrete actions: no-op and $\pm x, \pm y$ thrust), with the *identical* interface to the baseline so the two models are directly comparable. The headline task is a single thrusted agent reaching a goal under momentum and friction; crucially, *velocity is unobservable from a single frame*, so a faithful state must infer it. The environment also supports a multi-body variant (elastic collisions, depth-ordered occlusion) and a *direct-control* variant (the action sets velocity directly), the latter being the action-driven regime in which label-free action discovery is well-posed (Section 3.3).

3.2 The world model: a discrete-latent RSSM

The Vision and Memory modules are fused into one Recurrent State-Space Model. The state has a deterministic recurrent part h_t and a *categorical* stochastic part z_t of shape (CAT×CLASSES); we use 24×24. With encoder $e_t = \mathrm{enc}(o_t)$:

$$h_t = \mathrm{GRU}\big([z_{t-1}, a_{t-1}],\, h_{t-1}\big), \qquad \text{prior } \hat{z}_t \sim p_\theta(z_t \mid h_t), \qquad \text{posterior } z_t \sim q_\theta(z_t \mid h_t, e_t). \tag{1}$$

Sampling uses the straight-through estimator with a 1% uniform mixture (*unimix*). The feature $\phi_t = [h_t,\, \mathrm{flat}(z_t)]$ drives a decoder $\hat{o}_t$, a **reward** head, and a **continue** head. The loss is the DreamerV3 objective

$$\mathcal{L} = \underbrace{w \cdot (\hat{o}_t - o_t)^2}_{\text{recon (fg-weighted)}} + \mathrm{CE}\big(r_{\mathrm{head}},\, \mathrm{twohot}(\mathrm{symlog}\, r_t)\big) + \mathrm{BCE}(c_{\mathrm{head}}, c_t) + \beta\, \mathrm{KL}_{\mathrm{bal}}\big(q \,\|\, p\big), \tag{2}$$

with KL balancing (mix of $\mathrm{KL}(\mathrm{sg}(q)\|p)$ and $\mathrm{KL}(q\|\mathrm{sg}(p))$) and *free bits* (clip each at 1 nat). Two design points proved essential at this scale (Section 4): **(i)** a *foreground-weighted* reconstruction so the small bright agent is not washed out by the dark background, and **(ii)** a two-hot value/reward support sized to the reward scale (255 buckets), without which the reward head learns only the mean.

3.3 Latent action model (Genie-style, label-free)

A separate model discovers a discrete action codebook from *frame pairs with no action labels*. An inverse model maps the latent change to a continuous code that is vector-quantized (VQ); a forward model predicts the next embedding from the current embedding plus the quantized code, so the code must carry the controllable change. We bias the inverse to quantize the latent *displacement* $e_{t+g} - e_t$, the change the action causes.

3.4 Control by imagination

We provide two controllers, both trained *without environment interaction.* **(1)** An actor-critic learned purely in imagination (DreamerV3): roll the prior forward, predict reward with the model's own head, bootstrap λ-returns with a slow target critic (two-hot value), and update the actor by REINFORCE-with-baseline plus entropy. **(2)** A **short-horizon model-predictive controller**: at each real step, encode to the posterior, score every action by a short (H=5) constant-action imagined rollout read out by a latent→position probe, take the best action, and replan. The short horizon is deliberate—long open-loop rollouts drift off the data manifold and the heads become unreliable there (Section 4).

3.5 JEPA track (decoder-free)

An online CNN encoder and an EMA target encoder, with an action-conditioned predictor trained to match the target embedding of the next frame (cosine loss) plus a VICReg variance/covariance term to prevent collapse—no pixels decoded.

4 Experiments

Setup. Everything trains from scratch on an Apple M-series laptop GPU (MPS): ∼400 episodes of random-policy rollouts; the RSSM for ∼2k gradient steps; the latent action model and JEPA track on the same rollouts; the controller needs no environment interaction. The world model is ∼7M parameters.

State faithfulness. A small probe on the frozen latent recovers the agent's *position* at $R^2 = 0.96$ (the baseline reported 0.97) and its *velocity*—which a single frame cannot show, and which the baseline never probed—at $R^2 = 0.55$. The latent is a genuine simulator state, not merely an appearance code (Table 1). Figure 1a shows that, given a single real frame and an action sequence, the model hallucinates the agent's trajectory open-loop.

Control. The imagination MPC controller reaches the goal **∼90%** of episodes (mean closest distance 0.019) versus 25% for a random policy (closest 0.183), well above the 2018 baseline's 39%. *Diagnosis of what made this work:* we measured how often a H-step imagined rollout identifies the goal-ward action—70% at H=5 but decaying to 55% at H=25 as imagination drifts out-of-distribution (the prior–posterior KL is ∼6 nats). A 25-step CEM plan therefore scored near chance and *underperformed random*; the short-horizon replanner fixes it.

Long-horizon drift. Open-loop dream pixel-MSE rises from ∼410 at horizon 5 to ∼720 at horizon 60 (Figure 1b)—the same exposure-bias drift flagged across the video-world-model literature, reproduced locally.

Label-free action discovery. This is the hardest capability and we report it precisely. On the momentum task the discovered codes sit at chance: the next frame is mostly predictable from velocity, so the action is a tiny residual and the VQ code is starved. Moving to the action-driven (direct-control) regime—where we verified displacement equals action 100% of the time—a *supervised* probe decodes the executed action from the latent transition at **87%** (chance 20%), but fully *unsupervised* VQ clustering recovers only **0.37 purity** (Figure 2). The 87→37 gap is the result: the latent displacement is position-dependent, so the same action looks different at different places and the codebook latches onto that nuisance variation. This is the genuine difficulty of unsupervised latent-action learning, not a bug.

Table 1: Head-to-head: 2018 V-M-C baseline vs. WM2 (2026), identical interface, both trained from scratch on a laptop. "–" = not measured by that system.

Capability	**2018 baseline**	**wm2 (2026)**
latent → position R^2	∼0.97	**0.96**
latent → velocity R^2	–	**0.55**
goal-reach success (no env interaction)	0.39 vs 0.05 rand	**0.90 vs 0.25 rand**
open-loop dream tracks agent	yes	yes
JEPA decoder-free predictor	–	cos 0.95, no collapse
long-horizon dream drift (MSE, $h5 \to h60$)	rises	410 → 722
label-free action ↔ true action	–	0.37 purity (0.87 supervised)
WorldScore-style mean	–	**0.82**

JEPA. The decoder-free predictor reaches cosine similarity ∼0.95 to the EMA-target next embedding with no representational collapse (per-dimension std ≈1), confirming the non-generative paradigm trains stably at this scale.

5 Discussion

The controlled comparison isolates what five years bought on this task. The discrete-latent RSSM is a strictly more faithful *state*: it encodes the hidden velocity, enabling a controller that beats the baseline without ever touching the environment. But the 2026 stack also pays for its generality. Two findings recur as small-scale echoes of frontier open problems. First, **long-horizon consistency**: imagined rollouts drift, and because the heads were trained on the posterior manifold they become unreliable on far-future prior states—so control must stay short-horizon and replan. Second, **unsupervised action learning**: the action is plainly present in the representation (87% supervised) yet hard to recover without labels (37% unsupervised), because the controllable signal is entangled with position. We also found that a purely *learned*-reward actor-critic underperformed a probe-guided planner here—the reward head is the weak link at this scale—so we deploy the planner and report the actor-critic honestly.

Engineering lessons. Three bugs, each a real phenomenon, stood between "it runs" and "it learns": (i) plain pixel-MSE washes out small objects, so foreground weighting is essential (and the goal must be small enough not to dominate it); (ii) a two-hot value support mismatched to the reward scale lets the head learn only the mean; and (iii) a lazy compressed-array loader re-decompressed 495 MB per sample, a 700× slowdown fixed by materializing once. These are documented in the accompanying findings so the next builder avoids them.

Limitations. The domain is a 2(.5)D toy; "physical faithfulness" is measured by latent probes, not a physics benchmark; the latent is small; and the label-free action result is partial. The point is the *architecture* and the *measured 2018→2026 delta*, reproducible on a laptop, not a frontier system.

6 Conclusion

We rebuilt the V-M-C skeleton with the 2026 frontier's techniques and benchmarked the two from scratch on a laptop. The modern, discrete-latent, imagination-based model is a more faithful state and a markedly better controller, while honestly inheriting the field's open problems—long-horizon consistency and unsupervised action discovery—in miniature. We hope the code, artifacts, and from-zero guide make the 2026 frontier legible to readers with no prior background.

References

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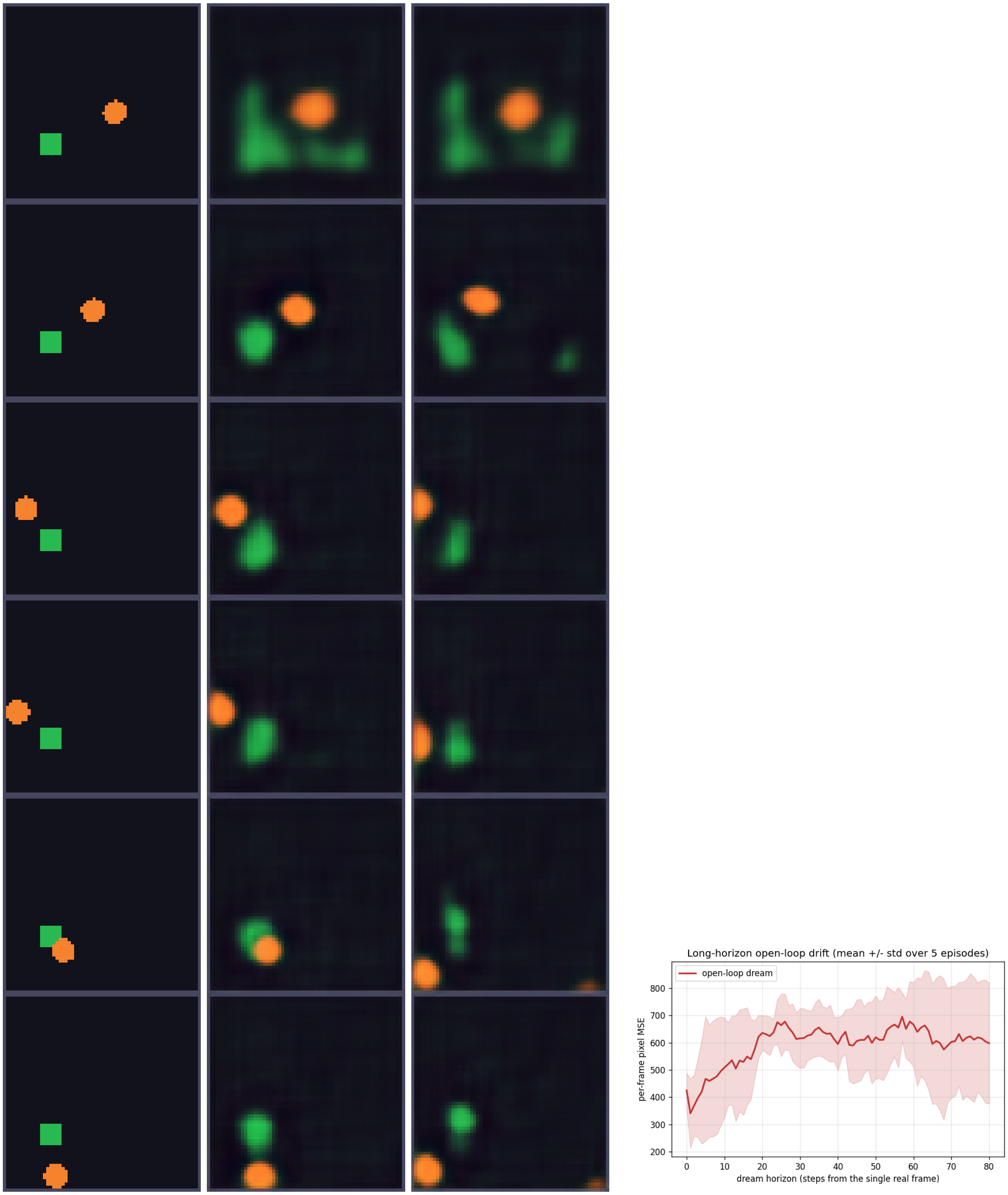

(a) real | reconstruction | open-loop dream

(b) long-horizon drift

Figure 1: **Left:** from one real frame and an action sequence the model dreams the agent's trajectory (middle = reconstruction ceiling, right = open-loop dream). **Right:** open-loop dream pixel-MSE vs. horizon (mean±std).

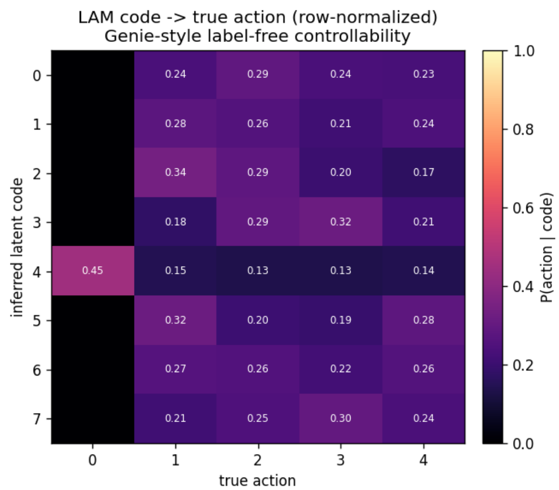

Figure 2: Latent action model: discovered code vs. true action (row-normalized) in the action-driven regime. A clean permutation would mean perfect label-free recovery; we obtain partial alignment (0.37 purity), while a *supervised* probe on the same features reaches 87%—quantifying the unsupervised gap.